

The empirical recurrence for OEIS A316126, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A316126 records “Empirical recurrence of order 67” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 71$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{67} = 55296 \neq 0$, so no further tail satisfies anything shorter; any order-67 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A316126 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 1, 2, 3, 5, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

2, 45, 197, 1112, 6562, 36480, 204465, 1170851, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Jun 24 2018 16:41:51, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 71$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 67: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 67.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 142$ exact terms from $n = 3$ on, it returns order 67 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	5	c_{18}	356852	c_{35}	241873542	c_{52}	−266728193
c_2	3	c_{19}	1083414	c_{36}	−49895168	c_{53}	−288646537
c_3	40	c_{20}	2504994	c_{37}	−1280553	c_{54}	372865888
c_4	−152	c_{21}	2062226	c_{38}	434045097	c_{55}	175344457
c_5	−217	c_{22}	2980015	c_{39}	−40017165	c_{56}	−145746744
c_6	−642	c_{23}	1345216	c_{40}	−313265316	c_{57}	2235118
c_7	889	c_{24}	1304101	c_{41}	322491519	c_{58}	43809990
c_8	3804	c_{25}	6490305	c_{42}	−544432912	c_{59}	−30745878
c_9	4210	c_{26}	−17540502	c_{43}	−1172629244	c_{60}	−4507958
c_{10}	5334	c_{27}	−37183906	c_{44}	−440111765	c_{61}	8234756
c_{11}	−7139	c_{28}	−35331829	c_{45}	−342655034	c_{62}	−2140968
c_{12}	1181	c_{29}	−63773680	c_{46}	−771273670	c_{63}	−897680
c_{13}	−8541	c_{30}	−40330314	c_{47}	−65329772	c_{64}	674080
c_{14}	−75593	c_{31}	80487404	c_{48}	527200716	c_{65}	40128
c_{15}	−213808	c_{32}	139417843	c_{49}	−429275472	c_{66}	−42240
c_{16}	−287128	c_{33}	36029411	c_{50}	−724760	c_{67}	55296
c_{17}	−318199	c_{34}	149068585	c_{51}	647932141		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A316126 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{67} - c_1 x^{66} - \dots - c_{67}$. Its constant term is $-c_{67} = -55296$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 67 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 67, so $\deg q = 67$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 67, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 67, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 55296.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-67 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A316126.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.